Architecture & Trade-offs

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The core has one primitive: `balance(account, value_date_upto, as_of_posting_day)`. It sums an append-only log where every record has two dates: the day it was posted and the day it is effective. There is no running balance; holds, fees and interest are all queries over the log. This is what makes the core correct when entries arrive backdated, and what makes it slow. The four sections below say what that costs at scale, what it means in a UAE bank, how authorizations end, and what I left out.

1 Append-only at scale

What breaks first: the end-of-day close

At every close I re-check every past day for every account, because a backdated entry can turn a closed day negative. Each check scans the whole log. One close costs about *days × accounts × entries*, and entries grow with *days × accounts*, so the cost is quadratic in both. At 100× accounts that is about 10,000× the work, and the close is the only job with a hard deadline. Nothing breaks; it stops finishing in time.

Where state grows without limit

The three logs (entries, authorizations, errors) grow by design; they are the audit record. The set of seen event ids grows without a plan. Every lookup is a linear scan, and one global sequence number ties all accounts together, so only one writer can run.

> **Cheapest fix: close the period**
> Set a horizon *H*, the last closed business day. Refuse any entry with a value date before *H* (one comparison in `append_entry`). At each close, save one snapshot per account: balance, holds and fee flags as of *H*. After that, a balance is the snapshot plus a scan of entries after *H*, and the close re-checks only the days since *H*. Audit does not change, because the snapshot is built from the log. Two things stay open: the log itself (five-year retention) and a sequence per account so accounts can run in parallel.

2 Value-dated entries in production

A backdated entry changes balances for days that are already closed. In Part 1, one backdated debit changed three closed days, added three fees in one close and declined an authorization. Its reversal changed the same days again and left the fees in place. In a UAE-licensed bank, each of these has an owner.

Customers and complaints	Statements already sent are now wrong, and a fee shows up for a day the customer saw as positive. Under the CBUAE Consumer Protection Standards (Circular 8/2020, Art. 8) a complaint is acknowledged within 2 business days, answered within 30, and kept on file for five years. Re-assessed fees will be the main source of these complaints.
Fees	Overdraft charges fall under the CBUAE retail fee caps (Regulations 29/2011, Annexure 2). A new fee, or a change above 5%, needs CBUAE approval first. One backdated debit turning into three charges is a fee behaviour compliance has to sign off and the customer agreement has to describe.
Finance	Interest is recomputed from the restated history. A backdated entry across a month-end changes income already reported, and the general ledger needs reversing journals the core does not produce.
Audit and AML	Every record has both dates, so “what did we know on Day 5” is a query, and the log meets the five-year retention in Federal Decree-Law No. 10 of 2025. What is missing is who backdated the entry and why.

> **The one control I would add before go-live: a backdating horizon with maker-checker**
> The core refuses any entry with a value date before the last closed day. Anything that has to go behind it goes to an adjustments queue, where a second person approves it with a reason code, and the approval is added to the log. This limits how much history can change, gives the examiner the who-and-why trail, and makes automatic fee re-assessment safe to run.

3 Authorization lifecycle

In the model an authorization is a decision record; its state (declined, active, settled) comes from whether a settlement references it. The only full ending I implemented is a matching settlement. The other endings in the model come first, then the ones production needs and the model does not have.

References: CBUAE Consumer Protection Standards (Circular 8/2020) Art. 8 · CBUAE Regulations 29/2011 Annexure 2 (2018) · Federal Decree-Law No. 10 of 2025 · Visa/Mastercard hold timeframes. Drafted with Claude, as disclosed in the Part 1 WORKLOG.

ENDING	REAL-WORLD CASE	WHAT I WOULD MANDATE
In the model		
Declined at request today: recorded, never revisited	Not enough available funds; in practice also fraud rules.	Final, with a reason code. A later backdated credit does not re-open it.
Duplicate request today: rejected as an error	Network retry.	Return the original decision; no second hold.
Never settled today: the hold stays active for ever	Abandoned pre-authorization: hotel, car rental, fuel, cancelled order.	Expire per scheme rules (Visa 7 days, 31 for lodging and rental; Mastercard 7, 30 for pre-authorizations); release the hold, record the expiry.
Settled for a different amount today: partial releases the whole hold; over-settlement is flagged	Bill below the estimate; tips above it.	Partial releases the remainder unless multi-capture was flagged; over-capture only within the scheme's tolerance for the merchant type.
Settlement reversed later today: funds return, hold not reinstated	Refund or chargeback.	Keep as is. A refund is a new credit. A chargeback is its own workflow.
Not in the model, needed in production		
Reversed by the merchant today: no such event	Order cancelled before capture.	New event. Full reversal voids the hold, partial reduces it. Only the merchant can send it.
Late presentment or no authorization today: rejected (Part 1 criterion 4)	Stand-in processing, offline chip approval, force-post.	Post it, flagged, so the ledger matches money the scheme already settled, and send it to disputes. The remedy is a chargeback, not refusal.
Account frozen or closed today: not modelled	Fraud freeze, legal hold, dormancy, closure.	A freeze blocks new authorizations but keeps existing holds; closure releases them and sends late clearings to disputes.
Incremental authorization today: same id is treated as a duplicate	Hotel stay extended.	Amend the existing hold and reset its expiry.

4 What I cut and why

STORAGE AND SCALE

In-memory lists, linear scans, one global sequence (persistence was out of scope)
Quadratic close; one writer; a crash loses the idempotency set, so a replay can post twice.

Backdating with no limit (the exercise needs it)
Full-history restatement. Covered by the horizon in section 2.

MONEY MODEL

Single-sided entries, no double-entry, no transfers (none in the stream)
No debits-equal-credits check. Nothing reconciles to the general ledger.

Simple daily interest on positive balances, one capitalisation; no debit interest, tiers or Islamic profit model (spec wording)
Wrong for any real product. A restated accrual across a reporting period has no journal.

HALF_UP, rounded daily then summed (spec wording)
Has to match the customer agreement. The difference is fils.

Only an AED fee configured (the spec gives one)
An overdrawn BHD account raises an error instead of a charge.

FEES

No link from a fee to the entry that caused it, no fee reversal (the spec never says when to unwind; this is the failing test)
A reversed bank error leaves the customer paying. Fix: a trigger reference on each derived entry and a compensating fee reversal.

Debits never declined, no overdraft limit, flat fee with no cap (the spec gives none)
Unlimited overdraft and fee stacking; needs a limit per account and a cap per cycle.

AUTHORIZATIONS

No hold expiry, merchant reversal, incremental authorization or multi-capture (none in the stream)
Holds lock funds for ever; a merchant void, the most common real ending, cannot be recorded.

Approvals never revisited after a backdated debit (this is how card networks work)
The exposure sits with the bank. Nothing flags “approved on a balance that was later restated”.

No account lifecycle (out of scope)
Freeze and closure behaviour for holds is undefined.

TIME

Integer days, no calendar (the brief's days are abstract)
Weekends, holidays, cut-off times and Gulf-time day boundaries are undefined.

Future value dates refused (keeps “balance as of day d” clear)
Standing and scheduled payments cannot be recorded.

INPUT AND ERRORS

Structural faults stop the replay (right for a test harness)
One bad message halts production; needs a dead-letter queue.

Rejections are only logged (the brief asks for errors to be printed)
A real force-post never reaches disputes, and settled money is unaccounted for.

No operator, channel, reason code or reference on entries (none in the stream)
Audit knows what and when, not who or why. No reconciliation to scheme files.

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